

Hence $P = 15 [2\pi(0.080) \times 10 / 1] = 75 \text{ W}$

8. **Ans: B**

Tension of cord provides for the centripetal force.

$$T = \frac{mv^2}{r}$$

$$kx = \frac{mv^2}{r}$$

$$(42)x = \frac{(0.80)(3.2)^2}{0.72}$$

$$x = 0.27 \text{ m}$$

$$\text{Unstretched length} = 0.72 - 0.27 = 0.45 \text{ m}$$